$$\mathcal{C}(\tau) = \int \mathrm{d}\Delta \, J(\Delta) \left\{ \cos\left(\Delta\tau\right) - \mathrm{i} \tanh\left(\frac{\Delta}{2k_{\mathrm{B}}T}\right) \sin\left(\Delta\tau\right) \right\}. \tag{5.186}$$